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AI LITERACY
&
DESIGN THINKING

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THE QUESTION

WHERE DOES AI FIT?

Design thinking has five phases. Generative AI is now in the room for all of them. Which ones does the literature actually report?

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WHAT WE'LL COVER

1. **The gap**
   Why phase-level evidence is missing, and why it matters for course design.
2. **The review**
   Arksey and O'Malley's five stages; four databases; 17 empirical studies.
3. **Corpus**
   Who was studied, where, in what disciplines, and over which years.
4. **Finding one**
   Phase coverage is lopsided: Prototype dominates, Empathize trails badly.
5. **Finding two**
   Knowing and using AI are reported far more than evaluating and creating it.
6. **So what**
   What the map supports, what it cannot, and what to do next.

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QUESTIONS

RESEARCH QUESTION 1

What is the scope and nature of empirical research on AI literacy in design thinking contexts?

RESEARCH QUESTION 2

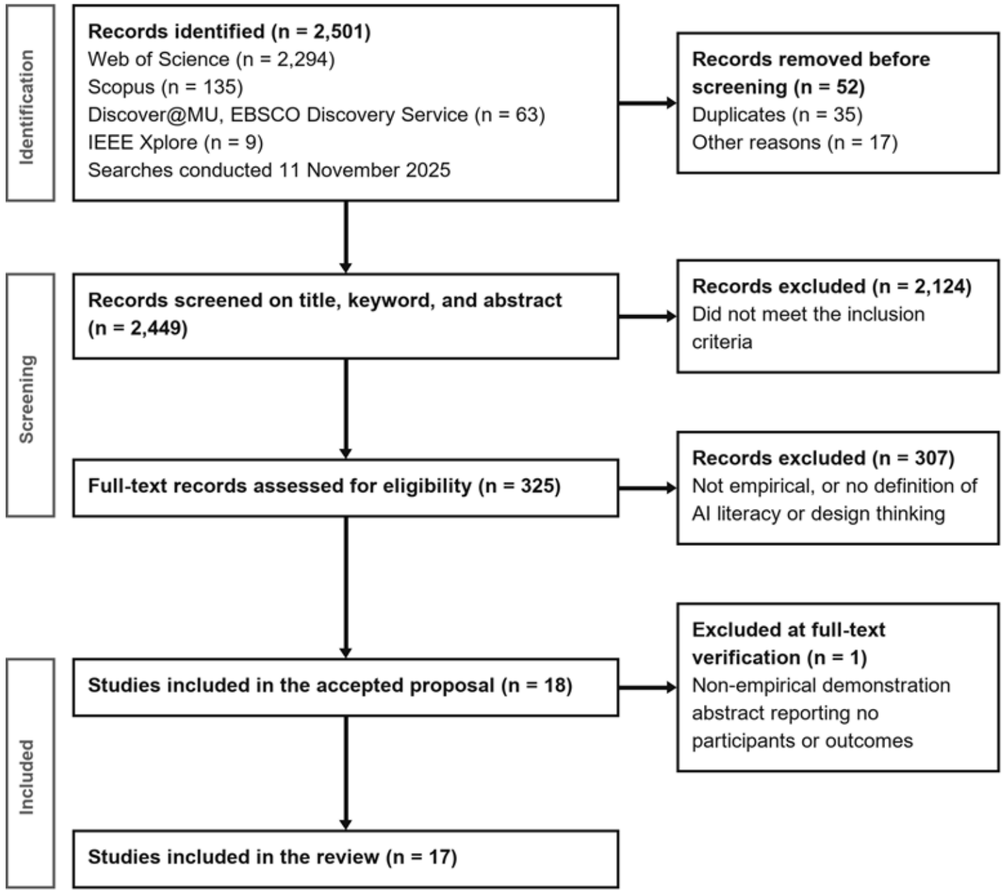
Which AI literacy dimensions are reported alongside which design thinking phases?

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METHOD

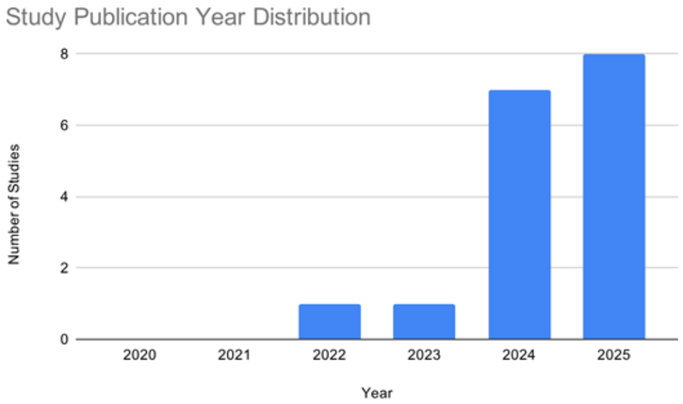
HOW WE ANSWERED THEM

Scoping review following Arksey and O'Malley (2005). Web of Science, Scopus, Discover@MU and IEEE Xplore, 2020–2025; searches closed 11 November 2025. 2,501 records screened down to 17.



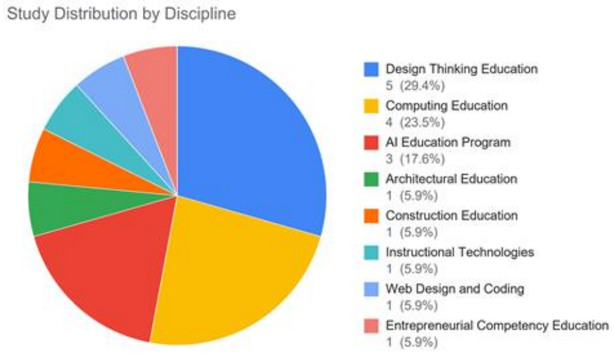
THE 17 STUDIES

WHEN



Fifteen of the 17 appeared in 2024 or 2025.

WHAT FIELD



Design thinking, AI and computing education lead; five fields have one study each.

WHERE



Led by the United States (5) and China (3); nine countries in all.

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PHASE COVERAGE

PROTOTYPE · 88%

Reported by 15 of 17 studies - where generative tools deliver something visible fastest.

THE MIDDLE

Test 59%, Ideate 53%, Define 47%. The middle of the process clusters tightly - about half the corpus.

EMPATHIZE · 29%

Five of 17 studies - a third as often as Prototype. The phase where learners are meant to understand the people they design for.

WHAT WE CANNOT SAY

Why. A review of reported coverage describes the distribution; it cannot explain it. Tool affordances, teaching priorities and reporting habits all plausibly contribute.

THE MAP ITSELF

Each row is one study. Orange marks a design thinking phase; blue an AI literacy dimension.

Study	Empathize	Define	Ideate	Prototype	Test	Direction	Know & Understand	Use & Apply	Evaluate & Create	AI Ethics
Solyst et al. (2023)				X		→	X	X	X	X
Dwivedi et al. (2024)		X	X	X	X	→			X	X
DiPaola et al. (2023)		X				→	X	X		
Han & Han (2025)				X	X	↔	X	X		X
Akman (2025)				X	X	→	X			X
Smith et al. (2025)	X		X			→	X			X
Wang et al. (2025)		X	X	X	X	↔	X	X	X	X
Kim & Lee (2023)				X		←	X	X		
Marçal et al. (2025)			X	X	X	↔	X	X	X	
Yue et al. (2025)		X	X	X	X	→	X	X	X	X
Imjai et al. (2024)	X	X		X	X	↔	X	X		
Rana et al. (2025)	X	X	X	X	X	↔	X	X	X	X
Yavuz et al. (2025)				X		→	X	X	X	X
Rong et al. (2024)			X	X		←	X	X	X	
Kim et al. (2025)	X	X	X	X	X	↔	X	X		X
Cheng et al. (2024)				X		→	X	X		X
Hsu & Hsu (2024)	X	X	X	X	X	↔	X	X	X	X

THE SHAPE OF IT

- AI literacy is reported where design work produces artifacts, and thins out where design work is about understanding people.
- Knowing and using AI: 94% and 82% of studies.
- Evaluating and creating AI: 53%.
- Codes sit at the study level, so Table 1 records co-occurrence, not where literacy developed inside a phase.

READING THE MAP

WHAT IT SHOWS

Which AI literacy dimensions are reported alongside which phases, across 17 studies, with every count traceable to a coded source.

ITS LIMITS

Codes sit at the study level, so the table records co-occurrence, not where literacy developed inside a phase.

FOR COURSE DESIGN

Early, human-centered phases are where AI support is thinnest. That is a design opportunity, not just a gap in the literature.

FOR RESEARCH

Report phases explicitly. Assess evaluative competence directly rather than inferring it from whether learners used a tool.

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